Evaluating ESM2 Generalisation for Binary Classification of Protease Cleavage Site Pairs

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Abstract

Proteases regulate biological processes by cleaving substrate proteins at specific amino acid positions. Predicting which sites are recognised by a given protease could support the study of disease mechanisms, enzyme function and therapeutic targets. This project investigated whether an ESM2 protein language model could predict whether a protease sequence would recognise a candidate cleavage site, based on MEROPS data. MEROPS provides positive cleavage records but does not provide experimentally confirmed negative pairs. Consequently, negative sampling and evaluation design were central parts of the study.

A sequential experimental workflow compared clustering strategies, negative sampling methods and frozen versus unfrozen ESM2 models. After selecting the clustering strategy, negative sampling method and model architecture through sequential experiments, follow up analyses examined model capacity, learning rate and validation split sensitivity. The later workflow also strengthened MEROPS family parsing and added saved split and pair manifests to improve reproducibility. The strongest completed configuration used an unfrozen ESM2 8M model, no clustering, cross family negatives and a learning rate of 10^{-5}. In the single completed run, the strongest configuration achieved a validation Matthews correlation (MCC) coefficient of 0.4681. Performance was substantially lower on held out proteases, with an MCC of 0.0241 for singleton proteases and 0.2549 for proteases represented by two to ten observations. These results indicate that ESM2 fine tuning learned a non-random signal, but did not generalise reliably to sparsely represented proteases.

Introduction

Proteases are enzymes that catalyse the hydrolysis of peptide bonds and regulate processes including protein processing, signalling, immune responses, apoptosis and tissue remodelling. Their biological effects depend not only on their abundance, but also on their activation, inhibition and localisation, as well as substrate recognition, including the amino acid and structural context surrounding the potential cleavage bond (López-Otín and Bond, 2008; Howng et al., 2021; Claushuis et al., 2023). Identifying these relationships experimentally is difficult because a protease can have many substrates, while a substrate can contain several possible cleavage positions(Rawlings, Barrett and Finn, 2014).

The MEROPS database organises proteases, substrates and known cleavage events using a hierarchical classification of catalytic classes, clans and families (Rawlings, Barrett and Finn, 2014). It provides a major source of observed protease cleavage relationships for computational modelling. However, the available records are unevenly distributed. Some proteases have thousands of observations, while others have only one or a small number. MEROPS also primarily records observed events rather than confirmed cases where cleavage did not occur.

Earlier cleavage prediction systems generally treated the task as protease specific site prediction. PROSPER used sequence profiles and predicted structural properties to predict cleavage sites for 24 proteases (Song et al., 2012). PROSPERous extended generic prediction to 90 proteases using sequence based scoring functions and logistic regression (Song et al., 2018). DeepCleave introduced convolutional neural networks and transfer learning, while Procleave incorporated sequence and structural information (Li et al., 2020a; Li et al., 2020b). These approaches demonstrated that cleavage specificity can be modelled, but their performance depends on data curation, sequence similarity, window length and the definition of negative examples (Li et al., 2019).

Protein language models provide an alternative way to represent amino acid sequences. These models are trained on large protein sequence collections and learn representations that encode biochemical, evolutionary and structural information (Rives et al., 2021). ESM2 extends this approach using Transformer models of different capacities. ESM2 representations have also supported structure prediction through ESMFold, indicating that they contain information about both local sequence patterns and broader protein organisation (Lin et al., 2023). More recent tools such as MPCutter apply protein language models directly to protease specific cleavage prediction (Wang et al., 2025).

This project used ESM2 to model the compatibility between two inputs: a full protease sequence and an eight residue cleavage site. This differs from a site only model because the protease identity is represented through its sequence rather than by training a separate classifier for each protease. The research question was:

Can an ESM2 protein language model distinguish observed MEROPS protease cleavage site pairs from constructed negative pairs, and does validation selected performance generalise to low sample held out proteases?

The completed experimental work reported here focused on binary classification. This formulation made the negative pair problem explicit and allowed different data preparation and modelling choices to be compared using standard classification metrics.

Methods

Data preparation

The processed MEROPS derived dataset contained 63,461 observed cleavage records across 912 protease codes and 48,287 unique cleavage site sequences. The final training file contained 61,436 records from 355 proteases. Two separate held out datasets contained proteases with limited representation:

Dataset	**Positive records**	**Protease codes**
Training data	61,436	355
Singleton held out set	127	127
Two to ten sample held out set	1,898	430

Physiological, nonphysiological and pathological cleavage records were retained. Each candidate cleavage site contained eight amino acids, corresponding to positions P4 to P4 prime around the cleavage bond. Protease inputs used the available full amino acid sequence.

All 557 protease codes in the two held out datasets were excluded from training and validation. The held out sets were not used to choose clustering, negative sampling, model architecture, learning rate, checkpoint or classification threshold.

Observed MEROPS pairs were assigned label 1. Since confirmed noncleavage records were unavailable, label 0 represented a constructed negative pair. Four negative strategies were examined:

1. **Scrambled:** residues within an observed cleavage site were shuffled.

2. **Random site:** a site was sampled from another record in the same data partition.
3. **Cross family:** a site was sampled from a protease belonging to another MEROPS family.
4. **Mixed:** scramble, random site and cross family sampling were alternated. Constructed negatives are not experimentally confirmed noncleavage events. They should therefore be interpreted as unobserved or deliberately mismatched combinations rather than true biological negatives.

Three clustering conditions were examined. The no clustering condition retained the full MEROPS derived training distribution. P1 clustering selected data according to biochemical groupings of the amino acid immediately before the cleavage bond, with the aim of reducing the dominance of common P1 residue patterns. Eight residue clustering grouped cleavage sites according to sequence similarity across the complete P4 to P4 prime window using a BLOSUM62 based similarity measure, with the aim of increasing sequence diversity within the selected data.

Sequential experimental design

Clustering was examined to determine whether reducing repeated or highly similar cleavage site patterns would prevent common motifs from dominating model training and improve generalisation to less represented cleavage sequences. The initial experiments therefore compared no clustering, P1 clustering and eight residue cleavage site clustering using scrambled negatives and frozen ESM2. The no clustering condition preserved the original data distribution, while the P1 and eight residue conditions tested progressively broader forms of sequence based redundancy reduction.

The follow up workflow refined the original pipeline by improving MEROPS family parsing, recording split and pair manifests, and strengthening reproducibility checks. Positive records were divided into training and validation partitions before negative pairs were generated independently within each partition. Duplicate binary pairs were removed using protease code, site sequence and label. Saved manifests confirmed that there was no exact positive pair or binary pair overlap between training and validation.

Model architecture

The main model was facebook/esm2_t6_8M_UR50D, which contains approximately eight million parameters. A frozen facebook/esm2_t12_35M_UR50D model was also evaluated to test whether additional pretrained capacity improved performance. The same ESM2 encoder processed the protease and cleavage site separately. Token embeddings were mean pooled using the attention mask and then L2 normalised. Four feature vectors were concatenated:

- the pooled protease embedding;
- the pooled site embedding;
- their absolute difference;
- their elementwise product.

This combined representation was passed through a multilayer perceptron containing a linear layer with 256 units, a rectified linear activation, dropout of 0.20 and a final linear output. Binary cross entropy with logits was used as the loss function.

ESM2 150M Follow Up

A limited ESM2 150M follow up was completed using the v2 sequential binary workflow. The model used facebook/esm2_t30_150M_UR50D, with the ESM encoder frozen and only the classifier head trained. The data setup remained fixed using no clustering and cross family negative sampling. The original code group random validation split was retained, with split seed 42, training seed 42 and pair generation seed 4242.

The model used a physical batch size of 1 and four gradient accumulation steps, giving an effective batch size of 4. Automatic mixed precision was enabled. The completed run was trained for nine epochs using a learning rate of 1×10^{-5}.

The frozen ESM2 150M model achieved a validation MCC of 0.0834, an AUROC of 0.5486, an F1 score of 0.6264 and an accuracy of 0.5479. The selected classification threshold was 0.40. This result was lower than the completed frozen ESM2 8M and 35M runs, which achieved validation MCC values of 0.1914 and 0.1961, respectively.

These results did not provide evidence that increasing the size of the frozen ESM2 backbone improved performance under the current setup. However, the 150M run used a different learning rate and completed fewer epochs than the 8M and 35M runs. It should therefore be interpreted as a limited exploratory follow up rather than a directly matched model capacity comparison.

A second frozen ESM2 150M run using a learning rate of 1×10^{-7} was attempted, but the Colab runtime disconnected before training was completed. The validation MCC remained at 0.0000 during the available epochs, and no final run summary was saved. This incomplete run was therefore excluded from the main results.

Model	LR	Epochs	Physical batch	Grad accum	Effective batch	Validation MCC	AUROC	Threshold
ESM2 8M frozen	10 -4	20	4	1	4	0.1914	0.6825	0.40
ESM2 35M frozen	10 -4	20	2	2	4	0.1961	0.6320	0.55
ESM2 150M frozen	10 -5	9	1	4	4	0.0834	0.5486	0.40

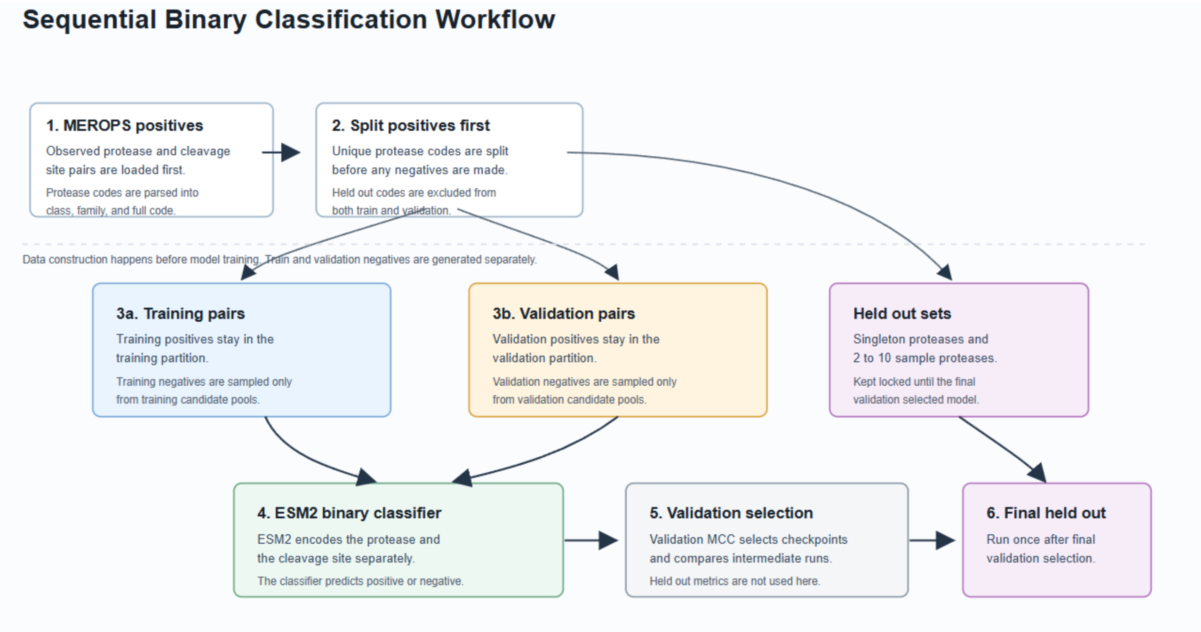
Training and evaluation

Models were trained for 20 epochs using AdamW, weight decay of 10^{-4}, gradient clipping at 1.0 and automatic mixed precision on a CUDA enabled Google Colab runtime (G4). The ESM2 8M models used a physical and effective batch size of four. The frozen ESM2 35M model used a physical batch size of two and two gradient accumulation steps, giving an effective batch size of four.

Frozen models used a learning rate of 10^{-4}. Unfrozen ESM2 8M models were tested at 10^{-7}, 10^{-6} and 10^{-5}. The principal validation split grouped records by protease code and used split seed 42. Training seed 42 and pair generation seed 4242 were used for the main runs.

MCC was selected as the primary model selection metric because it remains informative when positive and negative class proportions are unequal and its interpretation is less dependent on label prevalence than metrics such as accuracy or F1 score. Although the datasets were designed to be approximately balanced, duplicate pair removal produced small deviations from a one to one class ratio. MCC would also remain suitable if different positive to negative ratios were examined in later experiments. Precision, recall, F1 score, accuracy and area under the receiver operating characteristic curve were also recorded. Classification thresholds from 0.05 to 0.95 were examined in steps of 0.05. The checkpoint and threshold with the highest validation MCC were selected. Final held out negatives were generated after model selection using seed 4342.

Validation sensitivity was examined across seeds 42, 43 and 44 using code group random, family stratified code and family abundance stratified code splits. These comparisons were diagnostic and were not used to select a split based on whichever produced the highest MCC.



[**Contribution note to confirm:** exact personal authorship boundaries, adapted code and contributions from other project members should be checked before submission.]

Results

Initial sequential experiments

'No clustering' produced the strongest result in the first stage, although the differences between clustering conditions were modest.

Clustering strategy	Validation MCC	AUROC
No clustering	0.1945	0.6309
Eight residue clustering	0.1780	0.6259
P1 clustering	0.1547	0.6135

Using no clustering, cross family negative sampling produced the highest validation MCC among the four negative strategies.

Negative strategy	Validation MCC	AUROC
Cross family	0.2297	0.6495
Scramble	0.1945	0.6309
Random site	0.1907	0.6245
Mixed	0.1843	0.6075

Unfreezing ESM2 produced the largest improvement in the original sequential workflow. Validation MCC increased from 0.2297 with frozen ESM2 to 0.5329 with unfrozen ESM2. Validation AUROC increased from 0.6495 to 0.8300. This indicated that task specific adjustment of the ESM2 parameters was more effective than using fixed embeddings in this setting.

The original validation selected model achieved held out MCC values of 0.1183 for singleton proteases and 0.2656 for proteases represented by two to ten observations. These results were substantially lower than the validation MCC and motivated further investigation of model capacity, learning rate and validation split sensitivity.

Follow up experiments

Increasing frozen model capacity from ESM2 8M to ESM2 35M produced only a small change in validation MCC.

Frozen model	Validation MCC	AUROC	Threshold
ESM2 8M	0.1914	0.6285	0.40
ESM2 35M	0.1961	0.6320	0.55

This suggests that additional capacity alone did not materially improve performance when the ESM2 weights were fixed.

Learning rate had a larger effect when ESM2 8M was unfrozen.

Learning rate	Validation MCC	AUROC	F1	Best epoch
10^{-7}	0.1230	0.5607	0.6438	17
10^{-6}	0.3354	0.7341	0.6832	20
10^{-5}	**0.4681**	**0.8026**	**0.7354**	9

The selected model used no clustering, cross family negatives, unfrozen ESM2 8M and a learning rate of 10^{-5}. Its best validation MCC was reached at epoch 9 with a threshold of 0.10. Training loss continued to decrease after this point, while validation MCC did not consistently improve, which indicates increasing separation between training optimisation and validation performance.

A limited frozen ESM2 150M follow up was completed using a learning rate of 10^{-5} over nine epochs. It achieved a validation MCC of 0.0834 and an AUROC of 0.5486, both lower than the results from the completed frozen ESM2 8M and 35M models. The ESM2 150M configuration was therefore not selected for held out evaluation.

Validation sensitivity

Validation MCC varied according to both split strategy and random seed.

Validation strategy	Mean MCC	Standard deviation	Range
Code group random	0.2272	0.0729	0.1791 to 0.3111
Family stratified code	0.2296	0.0329	0.1983 to 0.2640
Family abundance stratified code	0.3915	0.1350	0.2406 to 0.5006

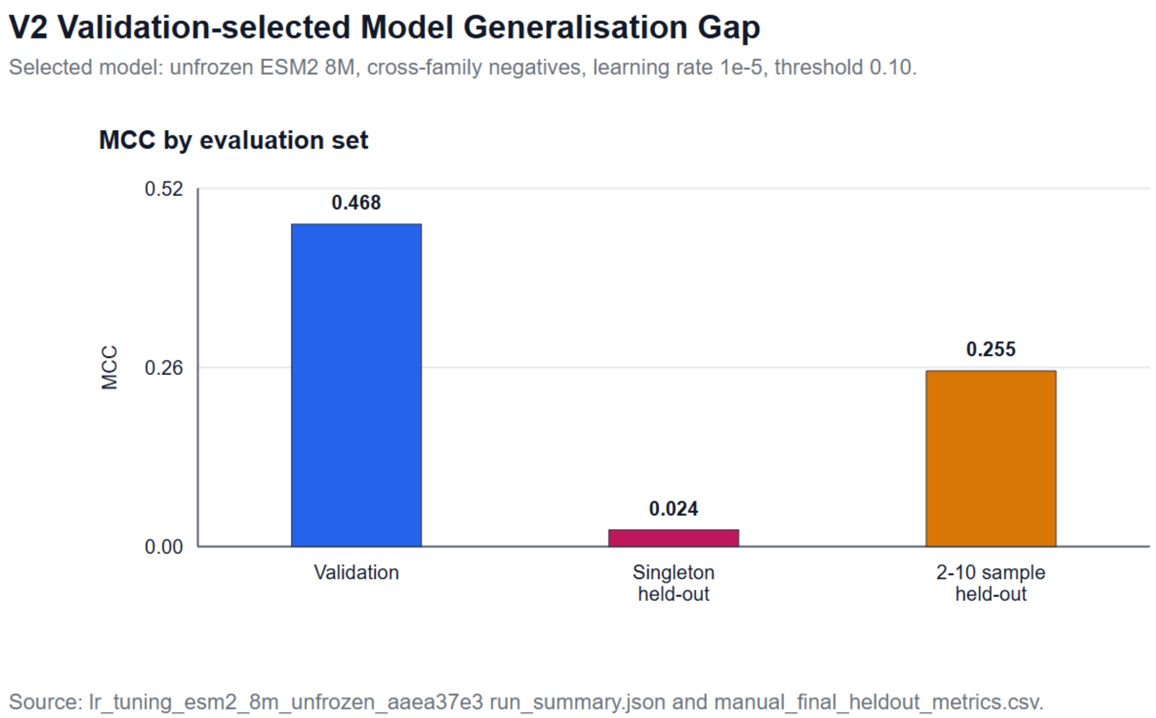
Family abundance stratification produced the highest mean MCC but also the greatest variation. These experiments used a frozen ESM2 8M model and were intended to measure sensitivity rather than identify a new best split. The results show that internal performance estimates depended strongly on which proteases were allocated to validation.

Final held out evaluation

The final validation selected model was evaluated once on the two held out sets using the validation selected threshold of 0.10.

Held out set	Pairs	MCC	Precision	Recall	F1	Accuracy	AUROC
Singleton proteases	254	0.0241	0.5098	0.6142	0.5571	0.5118	0.5435
Two to ten sample proteases	3,655	0.2549	0.5912	0.7171	0.6481	0.6227	0.6719

The singleton result was close to random classification. Performance was better for proteases represented by two to ten observations, but remained substantially below the validation result. The gap between validation MCC of 0.4681 and held out MCC values of 0.0241 and 0.2549 shows that validation performance did not provide a reliable estimate of generalisation to sparsely represented proteases.



Conclusions

This project developed a binary classification workflow for predicting compatibility between protease sequences and eight residue cleavage sites using ESM2. The main result was that fine tuning the ESM2 8M model produced a clear improvement over frozen embeddings during validation. Cross family negatives performed better than scrambled, random site and mixed negatives in the initial sequential comparison, while clustering did not improve the selected validation metric.

The follow up workflow strengthened the experimental design through improved MEROPS family parsing, saved split and pair manifests, partition specific negative generation and checks for overlap between training and validation. Under this workflow, the best completed configuration achieved a validation MCC of 0.4681 in a single training run. However, performance fell to 0.0241 on singleton proteases and 0.2549 on proteases represented by two to ten observations.

The results therefore support a limited conclusion. ESM2 fine tuning learned a useful signal from protease and cleavage site sequences, but this signal did not generalise robustly to data poor proteases. The stronger result on the two to ten sample set suggests that the model learned more than random noise. Nevertheless, the singleton result indicates that it cannot yet infer reliable cleavage specificity for a protease represented by a single observation.

Several limitations affect interpretation. The negative labels were constructed rather than experimentally confirmed, so some pairs labelled as negative may represent unknown positive cleavage relationships. MEROPS observations were also unevenly distributed across proteases and MEROPS families. The model may therefore have learned a mixture of genuine protease specificity, family level sequence patterns, dataset abundance effects and patterns introduced by the negative sampling procedure.

The main unfrozen ESM2 experiments were completed using a single training seed. The validation sensitivity analysis also showed that validation MCC varied across split strategies, meaning that validation performance should not be treated as a fully stable estimate of generalisation. Final held out performance was substantially lower than validation performance, particularly for singleton proteases, indicating that generalisation to proteases with very limited data remains weak.

Available computing resources materially restricted the experiments that could be completed. Colab disconnections and runtime instability became a greater problem as larger ESM2 models were attempted. This affected both the number of epochs that could be completed and the range of frozen and unfrozen models that could be compared. In particular, the completed ESM2 150M experiment was limited to a frozen encoder and nine epochs. Attempting to fine tune a model of this size would have required substantially more memory and training time and was not feasible within the available Colab resources.

A second frozen ESM2 150M run using a different learning rate was interrupted after approximately eight epochs when the runtime disconnected. As no final run summary was saved, this experiment could not be included as completed evidence. These interrupted runs also consumed paid Colab compute units without producing usable final outputs, which further limited the number of larger model experiments that could be attempted. The resulting model capacity comparison was therefore shaped partly by computational feasibility rather than scientific preference alone.

Further work should repeat the selected fine tuning configuration across multiple training seeds, test larger unfrozen ESM2 models using more stable or dedicated GPU resources, and develop validation splits that better represent low sample deployment. Experimentally confirmed negative cleavage relationships would provide a stronger basis for binary classification, although such data are difficult to obtain. Future evaluation should also consider protease family generalisation, substrate similarity and threshold calibration.

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